

BTM415 Game Theory

Introduction

Seungwon (Eugene) Jeong

KAIST BTM

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Logistics

- Instructor

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- ▶ Office: N22 #403
- ▶ Office Hours: Tue 4:00-5:00pm (by appointment)
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- Class Meetings

- ▶ Tue, Thu 10:30-11:45am, N22 #102

- Grading Policy

- ▶ 5 homework assignments (20%)
- ▶ midterm exam (30%)
- ▶ final group presentation (30%)
- ▶ attendance/participation (20%)

- Prerequisites: While there are no formal prerequisites, strong mathematical skills and a basic understanding of microeconomic theory would be helpful.
- Textbook: None is required but having one of 1 or 2 in the following is highly recommended. For auction theory, 6 is the standard reference.
 - ① Tadelis, Steven. *Game theory: an introduction*. Princeton university press, 2013.
 - ② Gibbons, Robert S. *Game theory for applied economists*. Princeton University Press, 1992.
 - ③ Osborne, Martin J., and Ariel Rubinstein. *A course in game theory*. MIT press, 1994.
 - ④ Osborne, Martin J. *An introduction to game theory*. Oxford university press, 2004.
 - ⑤ Binmore, Ken. *Playing for real: a text on game theory*. Oxford university press, 2007.
 - ⑥ Krishna, Vijay. *Auction theory*. Academic press, 2009.

Tentative Course Schedule

- Week 1 Introduction to game theory
- Week 2 Static games of complete information (Normal-form game)
- Week 3 Static games of complete information
- Week 4 Dynamic games of complete information (Extensive-form game)
- Week 5 Dynamic games of complete information (Subgame-perfect Nash equilibrium)
- Week 6 Static games of incomplete information (Bayesian Nash equilibrium)
- Week 7 Auction theory
- Week 8 Midterm Exam

Tentative Course Schedule (continued)

- Week 9 Mechanism design (Optimal mechanism)
- Week 10 Market Design (Matching theory)
- Week 11 Dynamic games of incomplete information (Perfect Bayesian equilibrium)
- Week 12 Dynamic games of incomplete information (Signaling game)
- Week 13 Cooperative game theory (Core and Shapley Value)
- Week 14 Bargaining
- Week 15 Additional topics and Review
- Week 16 Final Group Presentation

This schedule is subject to change if necessary.

Simple auction experiments

- Suppose you value the item at \$100. How much will you bid?
- First-price auction
- Second-Price Auction
- English Auction
- Dutch Auction

Simple auction experiments

Roughly, all expected revenues are equivalent (under certain conditions)
Formally, *Revenue Equivalence Theorem*

Math Review

- Given a random variable $X \sim F : [a, b] \rightarrow [0, 1]$ (pdf: $f(x)$),
- $E[X]$
- $E[g(X)]$ for some function g
- $E[X|X < x]$
- $\int_a^b f(x) g'(x) dx$ (Integration by parts)
- $\frac{d}{dx} f^{-1}(x)$
- change the order of integration
 - ▶ $\int_0^1 \int_0^x f(x, y) dy dx = \int_?^? \int_?^? f(x, y) dx dy$
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